

Biogeochemical cycle

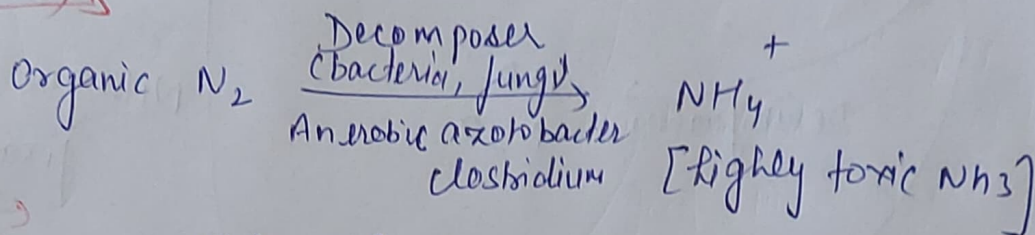
①

① Nitrogen Cycle [Gaseous Cycle]

In nitrogen cycle include

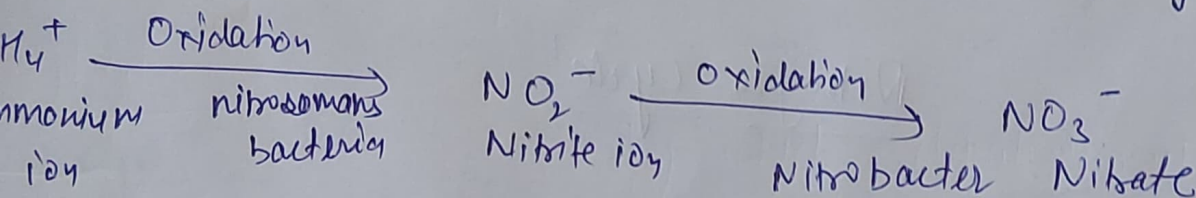
- Nitrogen fixation
- Ammonification
- Nitrification
- denitrification

Ammonification $\rightarrow$ Nitrogen fixation



Nitrification

It is a oxidation of ammonium ion to nitrite and subsequent oxidation of nitrite to nitrate, [by nitrifying bacteria]

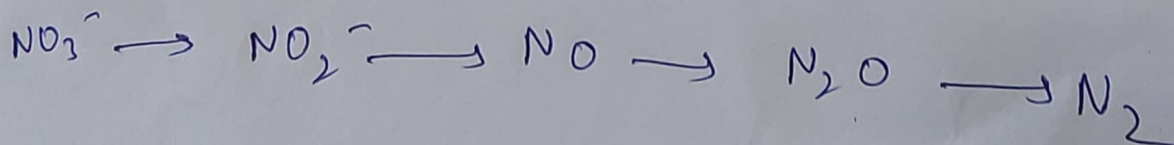


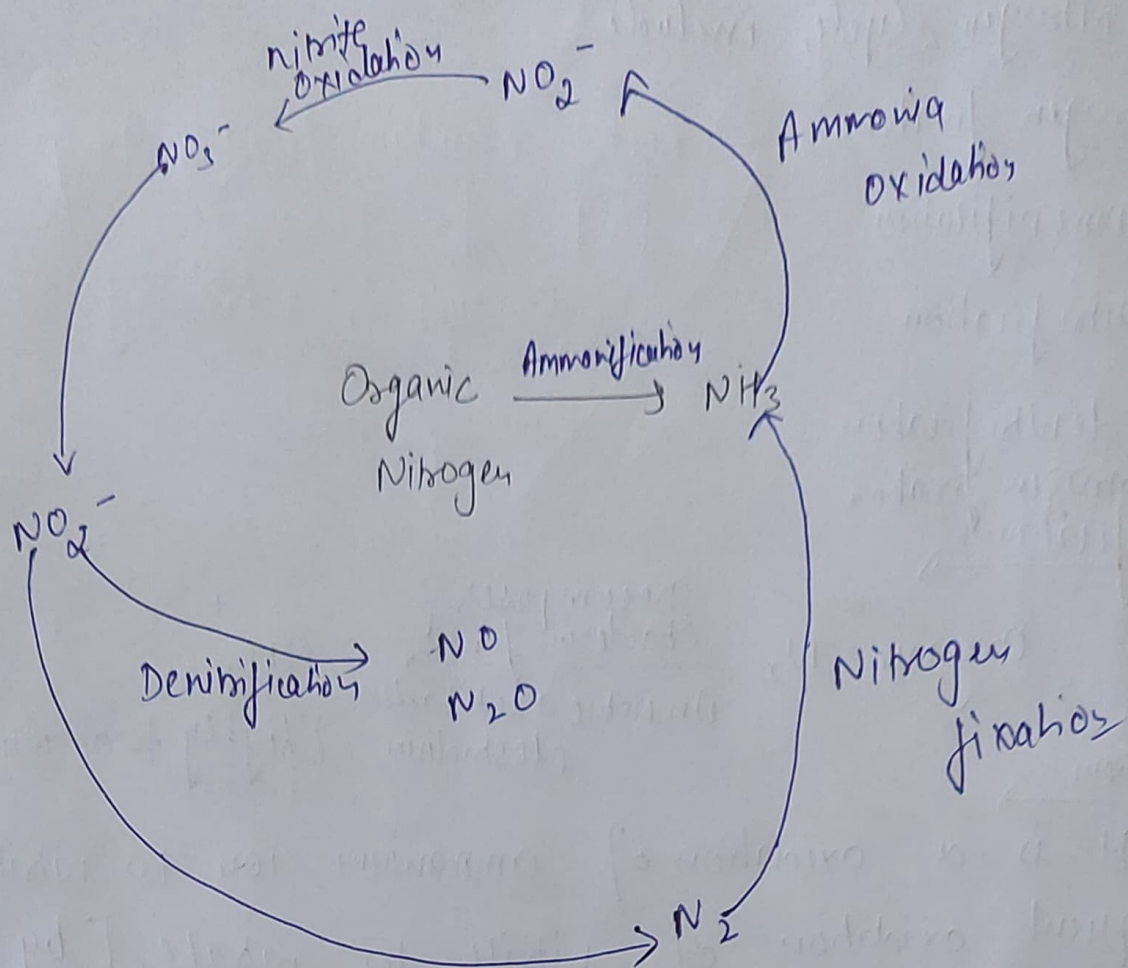
Denitrification

In which nitrate (NO_3^-) is converted back to Gaseous Nitrogen (N_2)

Anaerobic condition.

Bacteria involves *Paracoccus denitrificans*, *Thiobacillus denitrificans*, *Pseudomonas bacteria*





Nitrogen Cycle

Carbon Cycle

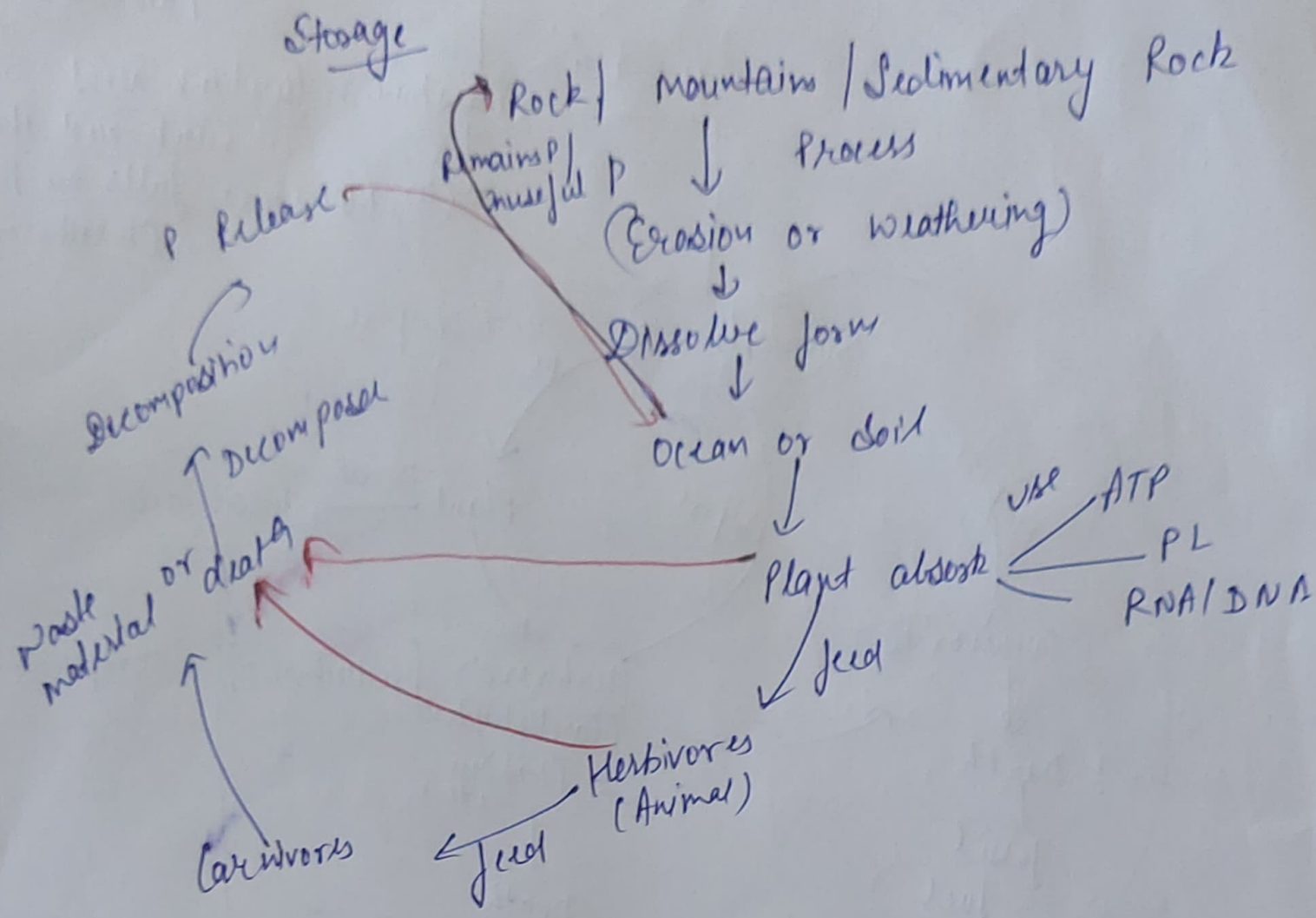
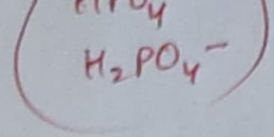
① Biological Cycle

Photosynthesis $\rightarrow \text{CO}_2$ taken in
Respiration $\rightarrow \text{CO}_2$ gives out

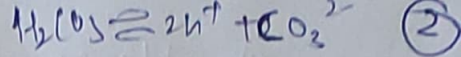
② Geological Cycle

Formation of carbonic acid, weathering, Sedimentation, Volcanic eruption

Phosphorus Cycle



Eutrophication → The process by which elevated nutrient levels leads to an overgrowth of algae and subsequent depletion of water oxygen levels is known as eutrophication.



from oceans

atm. O_2 +
Rainwater

Limestone

Carbonic acid
weak acid that
falls on Rock
(weathering)

(CO_2 Reservoir)

$H_2O + CO_2$

Atmospheric CO_2

Respire

PS

Plant

Death

Fossil fuel

Respire

Carnivores

Death

Fossil fuel

Herbivore

(Carbohydrate,
Protein,
etc.)

(production of C-compound)

Death

Fossil fuels

Burning of fossil fuel

Carbon Cycle

